APPENDIX I: Database search strategies

Medline (Ovid) search conducted on the 24/09/2020

#	Searches	Results
1	exp Diabetes, Gestational/	12 772
2	exp pregnancy in diabetics/	12 878
3	exp diabetes mellitus/ and exp pregnancy/	22 161
4	exp hyperglycemia/ and exp pregnancy/	2232
5	exp insulin resistance/ and exp pregnancy/	3047
6	(Gestation* adj3 (diabet* or DM or glucose intoleran* or insulin resistan*)).tw.	15 900
7	(Pregnan* adj3 (diabet* or DM or glucose intoleran* or insulin resistan*)).tw.	9783
8	(matern* adj3 (diabet* or DM or glucose intoleran* or insulin resistan*)).tw.	3810
9	GDM.tw.	7620
10	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9	37 673
11	exp regional health planning/	39 846
12	rural health/	23 422
13	suburban health/	469
14	rural population/	59 831
15	suburban population/	1903
16	hospitals, rural/	4874
17	exp rural health services/	12 946
18	suburban health services/	180
19	exp health services accessibility/	111 965

20	healthcare disparities/	17 243
21	rural*.tw.	141 141
22	remote*.tw.	77 137
23	regional.tw.	255 234
24	11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23	624 283
25	10 and 24	695
26	limit 25 to (english language and yr = “"1990 -Current”")	603

EMCare (Ovid) search conducted on the 24/09/2020

umber	Searches	Results
1	exp pregnancy diabetes mellitus/	10 548
2	exp diabetes mellitus/ and exp pregnancy/	7046
3	exp diabetes mellitus/ and exp pregnancy outcome/	2491
4	exp hyperglycemia/ and exp pregnancy/	670
5	exp hyperglycemia/ and exp pregnancy outcome/	225
6	exp insulin resistance/ and exp pregnancy/	814
7	exp insulin resistance/ and exp pregnancy outcome/	168
8	(Gestation* adj3 (diabet* or DM or glucose intoleran* or insulin resistan*)).tw.	6877

9	(Pregnan* adj3 (diabet* or DM or glucose intoleran* or insulin resistan*)).tw.	2755
10	(matern* adj3 (diabet* DM or glucose intoleran* or insulin resistan*)).tw.	95
11	GDM.tw.	3199
12	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11	15 005
13	rural health/	749
14	exp suburban population/	224
15	exp rural hospital/	106
16	rural health care/	4316
17	rural population/	12 641
18	health care access/	37 432
19	health care disparity/	6033
20	rural*.tw.	62 853
21	remote*.tw.	23 226
22	regional.tw.	80 487
23	urban rural difference/	2526
24	rural area/	26 905
25	13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24	203 446
26	12 and 25	479
27	limit 26 to (english language and yr = “"1990 -Current”")	466

CINAHL (EBSCOhost) search conducted on the 24/09/2020

#	Query	Limiters/Expanders	Last run via	Results
S38	S18 AND s36	Limiters - Published Date: 19900101-20201231; English Language Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	367
S37	S18 AND s36	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	374
S36	S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	215,767
S35	AB regional	Expanders—Apply equivalent subjects	Interface—EBSCOhost Research Databases	45,403

			Search modes—Boolean/Phrase	Search Screen—Advanced Search Database—CINAHL Complete	
	S34	TI regional	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	16,658
	S33	AB remote*	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	13,200
	S32	TI remote*	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	5,707
	S31	AB rural*	Expanders—Apply equivalent subjects	Interface—EBSCOhost Research Databases	43,826

			Search modes—Boolean/Phrase	Search Screen—Advanced Search Database—CINAHL Complete	
	S30	TI rural*	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	29,974
	S29	(MH "'Rural Health Centers"')	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	338
	S28	(MH "'Healthcare Disparities"')	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	11,881
	S27	(MH "'Health Services Accessibility+"')	Expanders—Apply equivalent subjects	Interface—EBSCOhost Research Databases	87,668

			Search modes—Boolean/Phrase	Search Screen—Advanced Search Database—CINAHL Complete	
	S26	(MH "'Services for Australian Rural and Remote Allied Health'")	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	5
	S25	(MH "'Rural Health Services'")	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	6,826
	S24	(MH "'Hospitals, Rural'")	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	2,899
	S23	(MH "'Suburban Population'")	Expanders—Apply equivalent subjects	Interface—EBSCOhost Research Databases	248

			Search modes—Boolean/Phrase	Search Screen—Advanced Search Database—CINAHL Complete	
	S22	(MH "'Rural Population"')	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	10,544
	S21	(MH "'Suburban Health"')	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	44
	S20	(MH "'Rural Health"')	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	6,916
	S19	(MH "'Rural Health Personnel"')	Expanders—Apply equivalent subjects	Interface—EBSCOhost Research Databases	657

			Search modes—Boolean/Phrase	Search Screen—Advanced Search Database—CINAHL Complete	
	S18	S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	14,705
	S17	AB (matern* n3 [(diabet* or DM or glucose intoleran* or insulin resistan*]))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	1,131
	S16	TI (matern* n3 [(diabet* or DM or glucose intoleran* or insulin resistan*]))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	417
	S15	AB GDM	Expanders—Apply equivalent subjects	Interface—EBSCOhost Research Databases	2,924

		Search modes—Boolean/Phrase	Search Screen—Advanced Search Database—CINAHL Complete	
S14	TI GDM	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	197
S13	AB (Pregnan* n3 [(diabet* or DM or glucose intoleran* or insulin resistan*]))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	2,701
S12	TI (Pregnan* n3 [(diabet* or DM or glucose intoleran* or insulin resistan*]))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	1,810
S11	AB (gestation* n3 [(diabet* OR DM OR “"glucose	Expanders—Apply equivalent subjects	Interface—EBSCOhost Research Databases	5,828

		intoleran*”" OR “"insulin resistan”"]))	Search modes—Boolean/Phrase	Search Screen—Advanced Search Database—CINAHL Complete	
	S10	TI (gestation* n3 [(diabet* OR DM OR “"glucose intoleran*”" OR “"insulin resistan”"]))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	4,838
	S9	((MH “"Insulin Resistance”") AND (MH “"Pregnancy Outcomes”"))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	78
	S8	((MH “"Insulin Resistance”") AND (MH “"Pregnancy+”"))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	752

S7	((MH “"Hyperglycemia”") AND (MH “"Pregnancy Outcomes”"))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	103
S6	((MH “"Hyperglycemia”") AND (MH “"Pregnancy+”"))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	491
S5	((MH “"Diabetes Education”") AND (MH “"Pregnancy Outcomes”"))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	23
S4	((MH “"Diabetes Mellitus+”") AND (MH “"Pregnancy Outcomes”"))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	1,792

S3	((MH “"Pregnancy+”") AND (MH “"Diabetes Education”"))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	145
S2	((MH “"Pregnancy+”") AND (MH “"Diabetes Mellitus+”"))	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	9,666
S1	MH “"pregnancy in diabetes+”"	Expanders—Apply equivalent subjects Search modes—Boolean/Phrase	Interface—EBSCOhost Research Databases Search Screen—Advanced Search Database—CINAHL Complete	9,074

EMBASE (Ovid) search conducted on the 24/09/2020

#	Searches	Results

1	exp pregnancy diabetes mellitus/	38 053
2	exp pregnancy/ and exp diabetes mellitus/	35 701
3	exp insulin resistance/ and exp pregnancy/	3659
4	exp hyperglycemia/ and exp pregnancy/	2747
5	(Gestation* adj3 (diabet* or DM or glucose intoleran* or insulin resistan*)).tw.	25 519
6	(Pregnan* adj3 (diabet* or DM or glucose intoleran* or insulin resistan*)).tw.	14 843
7	GDM.tw.	12 455
8	(matern* adj3 (diabet* or DM or glucose intoleran* or insulin resistan*)).tw.	5840
9	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8	63 665
10	regional health planning/	102 769
11	rural health/	1208
12	exp rural health care/	13 795
13	suburban health/	1101
14	exp rural population/	49 236
15	exp suburban population/	685
16	exp health care access/	63 203
17	exp health care disparity/	15 816
18	exp rural hospital/	248
19	rural*.tw.	175 617
20	remote*.tw.	99 230
21	regional.tw.	364 744
22	10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21	793 324
23	9 and 22	1473
24	limit 23 to (english language and yr="1990 -Current")	1363

Maternity and Infant Care (Ovid) search conducted on the 24/09/2020

#	Searches	Results
1	gdm.mp. [mp=abstract, heading word, title]	2217
2	(Gestation* adj3 (diabet* or DM or glucose intoleran* or insulin resistan* or hyperglyc*)).mp.	5267
3	(Pregnan* adj3 (diabet* or DM or glucose intoleran* or insulin resistan* or hyperglyc*)).mp.	2496
4	(matern* adj3 (diabet* or DM or glucose intoleran* or insulin resistan* or hyperglyc*)).mp.	989
5	rural*.mp. [mp=abstract, heading word, title]	5615
6	regional.mp. [mp=abstract, heading word, title]	4207
7	remote*.mp. [mp=abstract, heading word, title]	933
8	suburban.mp. [mp=abstract, heading word, title]	179
9	health service* access*.mp. [mp=abstract, heading word, title]	836
10	health* disparit*.mp. [mp=abstract, heading word, title]	239
11	1 or 2 or 3 or 4	6845
12	5 or 6 or 7 or 8 or 9 or 10	11 202
13	11 and 12	164

Cochrane Library search conducted on the 24/09/2020

ID	Search term	Search hits
1	MeSH descriptor: [Diabetes, Gestational] explode all trees	937
2	MeSH descriptor: [Pregnancy in Diabetics] explode all trees	318
3	[mh “"Diabetes Mellitus”"] AND [mh Pregnancy]	768
4	[mh Hyperglycemia] AND [mh Pregnancy]	72

5	[mh Pregnancy] AND [mh “"Insulin Resistance”"]	145
6	(gdm):ti,ab	1260
7	(gestation* NEAR/3 (diabet* OR DM OR “"glucose intolerance”" OR “"insulin resistance”")):ti,ab	2472
8	(Pregnan* NEAR/3 (diabet* OR DM OR “"glucose intolerance”" OR “"insulin resistance”")):ti,ab	1063
9	(matern* NEAR/3 (diabet* OR DM OR “"glucose intolerance”" OR “"insulin resistance”")):ti,ab	200
10	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9	3312
11	MeSH descriptor: [Rural Health] this term only	524
12	MeSH descriptor: [Suburban Health] this term only	18
13	MeSH descriptor: [Rural Population] this term only	1652
14	MeSH descriptor: [Suburban Population] this term only	33
15	MeSH descriptor: [Hospitals, Rural] explode all trees	34
16	MeSH descriptor: [Health Services Accessibility] explode all trees	934
17	MeSH descriptor: [Healthcare Disparities] this term only	180
18	MeSH descriptor: [Regional Health Planning] explode all trees	272
19	rural:ti,ab (Word variations have been searched)	8508
20	remote:ti,ab (Word variations have been searched)	5635
21	regional:ti,ab (Word variations have been searched)	40 911
22	#11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21	55 022

23	#10 AND #22 with Cochrane Library publication date Between Jan 1990 and Dec 2020	98

Rural and Remote Health (Informit) search conducted on the 24/09/2020

Search History

Set	Search Terms	Limits	Records
18	(GDM OR “"gestational diabetes”" OR Gestation* %3 diabet* OR gestation* %3 DM OR gestation* %3 “"glucose intoleran*”" OR gestation* %3 “"insulin resistan*”" OR gestation* %3 “"diabetes mellitus”" OR gestation* %3 @hyperglycemia OR gestation* %3 “"insulin resistan*”" OR matern* %3 diabet* OR matern* %3 DM OR matern* %3 “"glucose intoleran*”" OR matern* %3 “"insulin resistan*”" OR matern* %3 “"diabetes mellitus”" OR matern* %3 @hyperglycemia OR matern* %3 “"insulin resistan*”" OR matern* %3 diabet* OR matern* %3 DM OR matern* %3 “"glucose intoleran*”" OR matern* %3 “"insulin resistan*”" OR matern* %3 “"diabetes mellitus”" OR matern* %3 @hyperglycemia OR matern* %3 “"insulin resistan*”" OR pregnan* %3 diabet* OR pregnan* %3 DM OR pregnan* %3 “"glucose intoleran*”" OR pregnan* %3 “"insulin resistan*”" OR pregnan* %3 “"diabetes mellitus”" OR pregnan* %3 @hyperglycemia OR pregnan* %3 “"insulin resistan*”") AND (“"regional health* plan*”" OR “"rural health*”" OR “"suburban health*”" OR “"rural population*”" OR “"suburban population*”"		34

OR “"rural hospital*”" OR “"rural health* service*”" OR “"suburban health* service*”" OR “"health* service* access*”" OR “"health* disparit*”" OR rural* OR remote* OR regional*)

Aboriginal and Torres Strait Islander health bibliography (informit) search conducted on the 24/09/2020

Search Terms	Limits	Records
3	(GDM OR “"gestational diabetes”" OR Gestation* %3 diabet* OR gestation* %3 DM OR gestation* %3 “"glucose intoleran*”" OR gestation* %3 “"insulin resistan*”" OR gestation* %3 “"diabetes mellitus”" OR gestation* %3 @hyperglycemia OR gestation* %3 “"insulin resistan*”" OR matern* %3 diabet* OR matern* %3 DM OR matern* %3 “"glucose intoleran*”" OR matern* %3 “"insulin resistan*”" OR matern* %3 “"diabetes mellitus”" OR matern* %3 @hyperglycemia OR matern* %3 “"insulin resistan*”" OR matern* %3 diabet* OR matern* %3 DM OR matern* %3 “"glucose intoleran*”" OR matern* %3 “"insulin resistan*”" OR matern* %3 “"diabetes mellitus”" OR matern* %3 @hyperglycemia OR matern* %3 “"insulin resistan*”" OR pregnan* %3 diabet* OR pregnan* %3 DM OR pregnan* %3 “"glucose intoleran*”" OR pregnan* %3	82

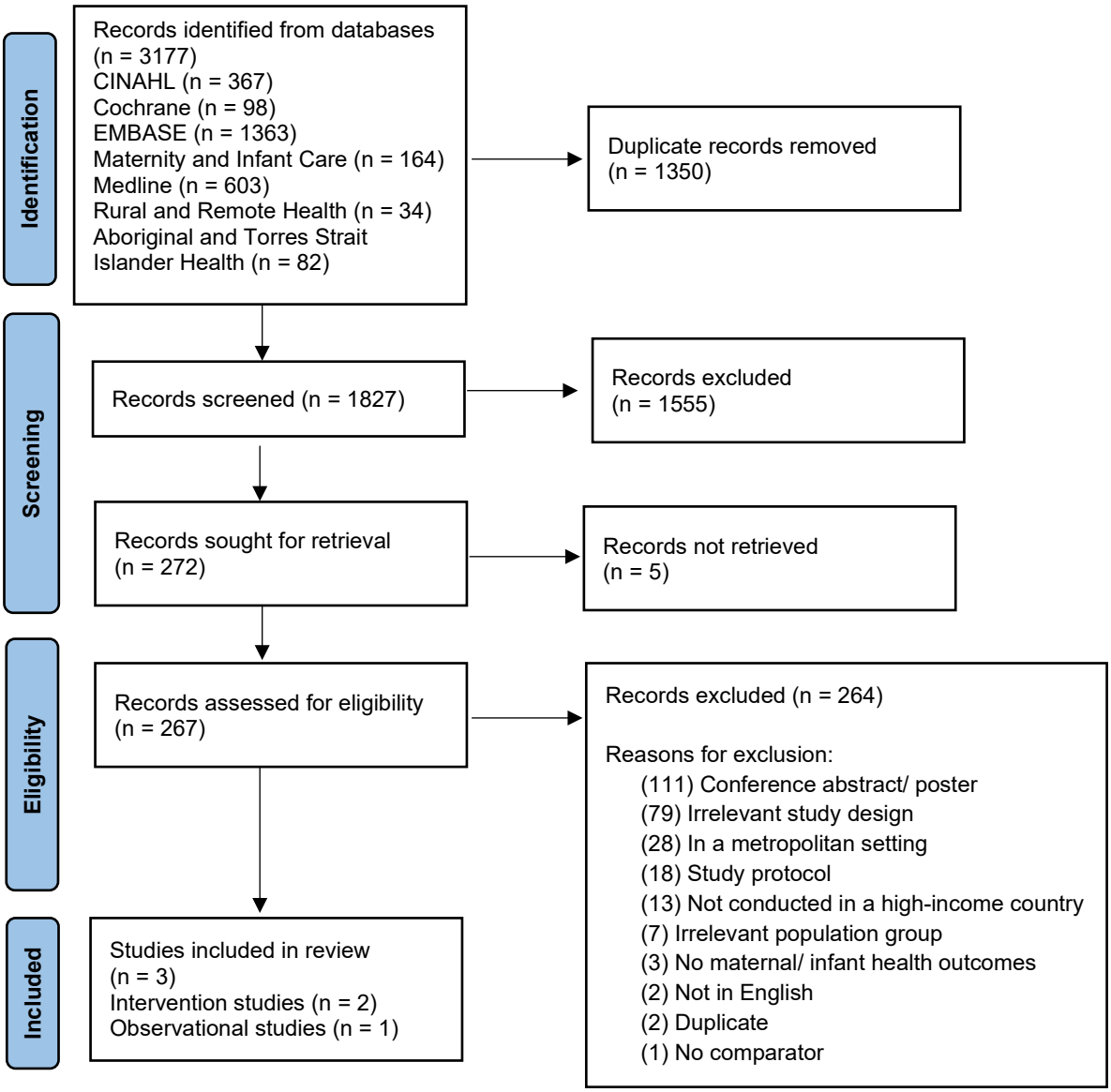
	“"insulin resistan*”" OR pregnan* %3 “"diabetes mellitus”" OR pregnan* %3 @hyperglycemia OR pregnan* %3 “"insulin resistan*”") AND (“"regional health* plan*”" OR “"rural health*”" OR “"suburban health*”" OR “"rural population*”" OR “"suburban population*”" OR “"rural hospital*”" OR “"rural health* service*”" OR “"suburban health* service*”" OR “"health* service* access*”" OR “"health* disparit*”" OR rural* OR remote* OR regional*)	

APPENDIX II: PRISMA diagram

APPENDIX III: primary outcomes, including maternal and infant physical and biochemical outcomes reported

Author, year	How measured?	Primary outcome/s identified?	Statistical tests used	Results	Conclusions
Murfet, 2014[46]	Data audit from maternity service records	Pooled adverse neonatal outcomes (macrosomia, hypoglycaemia, still	Adjusted relative risk (aRR) using Poisson regression adjusting for maternal age, pre-	All diabetes types Adjusted relative risk	A nurse practitioner-led model of care may reduce the frequency of adverse neonatal

Appendix II: PRISMA diagram



birth or neonatal death, respiratory distress syndrome, congenital abnormality, pre-term delivery, neural tube defect, birth injury)	existing diabetes, socioeconomic status and Asian or Indigenous ethnicity	(aRR) adverse neonatal outcomes 0.76 (95% CI 0.61–0.94) aRR adverse maternal outcomes 0.75, (95% CI 0.52–1.09)	outcomes in a rural setting, without increasing referrals to physicians.
Pooled adverse maternal outcomes (hypoglycaemia, loss of consciousness, diabetic ketoacidosis, metabolic complication, threatened abortion requiring cervical sutures, polyhydramnious, premature labour/placenta previa,		GDM group only aRR adverse neonatal outcomes 0.60 (95% CI 0.48–0.76) aRR adverse maternal outcomes 0.80 (95% CI 0.54–1.20)	

		pyelonephritis, emergency caesarean section/failure to progress, 2–4° tear, post-partum haemorrhage)			
Casey, 2019[45]	Sample from public hospital clinical records	Rates of neonatal hypoglycaemia	Chi-squared test	P value 0.89	No association was found between antenatal expression of colostrum and rates of neonatal hypoglycaemia or median blood glucose levels.
Weiderman, 1996[47]	Trained clinic staff from respective sites recorded	None identified	Not described for all outcome measures t-test used for comparison of birthweight	Apgar score, admission to neonatal intensive care unit, gestational age at delivery and method of delivery not significantly different	Small sample size limits ability to draw conclusions. Trends suggest that care can be provided to women with diet controlled gestational diabetes

Chi squared test used for comparison between groups of provider contact time	between groups – no values reported Birthweight *p* value 0.4	mellitus in a rural setting with comparable outcomes to a multidisciplinary approach.

APPENDIX IV

Secondary outcomes reported, including those relating to nutrition, medications, behaviour, exercise, psychology, service delivery and economic changes

Author, year	Nutrition	Medication	Behavioural	Exercise	Psychological/ mental health	Service delivery	Economic	Other
Murfet, 2014[46]	Diet/ exercise only Diet/ exercise and insulin	Diet/ exercise and insulin	-	Diet/ exercise only Diet/ exercise and insulin	-	Referral to physician for diabetes management Referral to diabetes educator for diabetes management	-	-

						Referral to dietitian for diabetes management Post-partum follow up		
Casey, 2019[45]	-	-	Expressed antenatal colostrum Baby formula fed Baby fully breastfed at discharge	-	-	Lactation consultant seen Days in special care nursery	-	-
Weiderman[47] 1996	Diet control only	Diet and insulin control	-	-	-	Frequency of visits Provider contact time	-	-

V: Quality assessment—ROBINS-I tool[43]

Studies	Robins-I tool domains							
	Domain 1 – confounding	Domain 2 – participant selection	Domain 3 – intervention classification	Domain 4 – intervention deviation	Domain 5 – missing data	Domain 6 – outcome measurement	Domain 7 – selection of reported result	Overall
Murfet[46] 2014	Serious	Low	Low	Moderate	Low	Moderate	Moderate	Serious
Casey[45] 2019	Moderate	Moderate	Moderate	Low	Low	Moderate	Moderate	Moderate
Weiderman[47] 1996	Serious	Low	Low	No information	No information	Moderate	Moderate	Serious